On the P_3-Hull Number

Hebert Coelho — UFG

Joint work with:

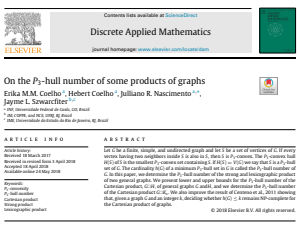
2019 — On the P_3-hull number of some products of graphs
Erika M. M. Coelho, Hebert Coelho, Julliano R. Nascimento, Jayme L. Szwarcfiter

2022 — P_3-convexity on graphs with diameter two: Computing hull and interval numbers
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2026 — Bounds on the P_3-Carathéodory number
Braully R. Silva, Erika M. M. Coelho, Hebert Coelho, Simone Dantas

Research discussion — Rio de Janeiro, October 2026

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On the P_3-hull number of some products of graphs

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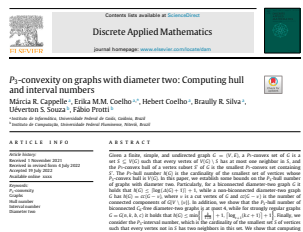
ABSTRACT

Let G be a finite, simple, and undirected graph and let S be a set of vertices of G. If every vertex v of G satisfies at least one of the following, then S is a P_3-hull set of G: (i) $v \in S$, or (ii) $v \notin S$ and the smallest P_3-convex set containing S, $H(S) = S \cup \{v\}$, we say that S is a P_3-hull set of G. The cardinality $h(G)$ of a minimum P_3-hull set in G is called the P_3-hull number of G. In this paper, we determine the P_3-hull number of the strong and lexicographic product of two general graphs. We present lower and upper bounds for the P_3-hull number of the Cartesian product, $G \square H$, of general graphs G and H, and we determine the P_3-hull number of the Cartesian product $G \square K_r$. We also improve the result of Centeno et al. 2011 showing that, given a graph G and an integer k, deciding whether $h(G) \geq k$ remains NP-complete for the Cartesian product of graphs.

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1. Introduction

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P_3-convexity on graphs with diameter two: Computing hull and interval numbers

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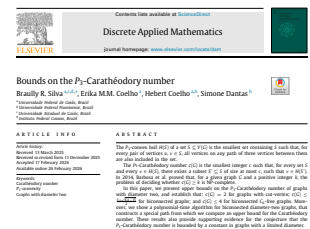
Keywords:
P_3-convexity
Graphs
Hull number
Interval number
Diameter two

ABSTRACT

Given a finite, simple, and undirected graph $G = (V, E)$, a P_3-convex set of G is a set $S \subseteq V(G)$ such that every vertex of $V(G) \setminus S$ has at most one neighbor in S, and the P_3-convex hull of a vertex subset S' of G is the smallest P_3-convex set containing S'. The P_3-hull number $h(G)$ is the cardinality of the smallest set of vertices whose P_3-convex hull is $V(G)$. In this paper, we establish some bounds on the P_3-hull number of the Cartesian product of graphs. We show that for every connected graph G of order n, it holds that $h(G \square K_2) \leq \lceil \log(\Delta(G) + 1) \rceil + 1$, while a non-biconnected diameter-two graph G has $h(G) = \alpha(\overline{G}) - \kappa_1$, where κ is a cut vertex of G and $\alpha(\overline{G} - \kappa)$ is the number of connected components of $\overline{G[V \setminus \{\kappa\}]}$. In addition, we show that the P_3-hull number of biconnected C_4-free diameter-two graphs is at most 4, while for strongly regular graphs $G = G(n, k, \lambda, t)$ it holds that $h(G) \leq \min\left\{\frac{n}{k+1}\left(\log_{\frac{k}{t}}(k(t-1)+1)+1\right), 4\right\}$. Finally, we consider the P_3-interval number, which is the cardinality of the smallest set S of vertices such that every vertex not in S has two neighbors in this set. We show that computing the P_3-interval number on diameter-two split graphs is NP-complete.

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Bounds on the P_3-Carathéodory number

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ABSTRACT

The P_3-convex hull $H(S)$ of a set $S \subseteq V(G)$ is the smallest set containing S such that, for every pair of vertices $u, v \in S$, all vertices on any path of three vertices between them are also included in the set.

The P_3-Carathéodory number $c(G)$ is the smallest integer ℓ such that, for every set S and every $v \in H(S)$, there exists a subset $S' \subseteq S$ of size at most ℓ, such that $v \in H(S')$. In 2014, Barbosa et al. proved that, for a given graph G and a positive integer k, the problem of deciding whether $c(G) \leq k$ is NP-complete.

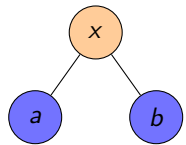
In this paper, we present upper bounds on the P_3-Carathéodory number of graphs with diameter two, and establish that: $c(G) = 2$ for graphs with cut-vertex; $c(G) \leq \frac{\Delta(\overline{G})+2}{2}$ for biconnected graphs; and $c(G) \geq 4$ for biconnected C_4-free graphs. Moreover, we chose a polynomial-time algorithm for biconnected diameter-two graphs, that constructs a special path from which we compute an upper bound for the Carathéodory number. These results also provide suggesting evidence for the conjecture that the P_3-Carathéodory number is bounded by a constant for graphs with diameter two.

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2026

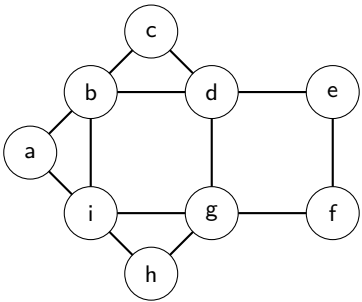
Motivation: activation and propagation

- Initially, some vertices are active.
- An inactive vertex becomes active once it has at least two active neighbours.
- Active vertices never become inactive.
- What is the smallest seed set that activates every vertex?



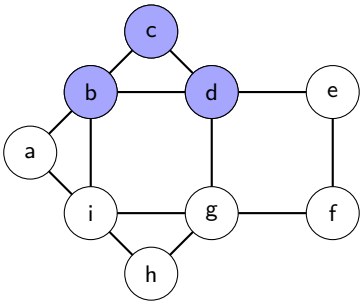
two active neighbours

P_3-Convexity: Convex Sets



A set $S \subseteq V(G)$ is P_3-**convex** if every vertex outside S has at most one neighbor in S.

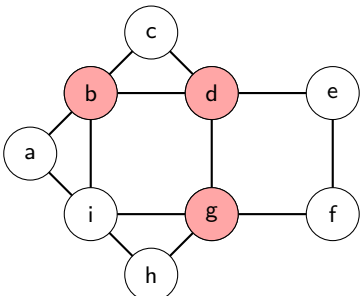
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- $S_1 = \{b, c, d\}$ is convex.

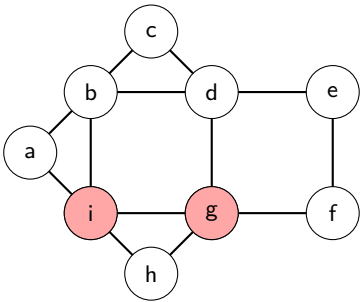
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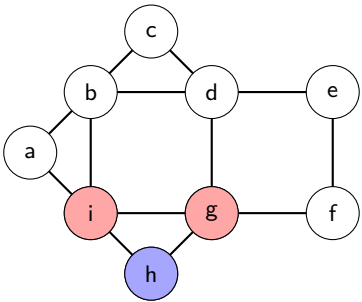
- $S_1 = \{b, c, d\}$ is convex.
- $S_2 = \{b, d, g\}$ is not convex: c and i have two neighbors in S_2.

P_3-Convexity: Interval



The P_3-**interval** $I[u, v]$ consists of u, v, and all internal vertices of P_3 paths with endpoints u, v.

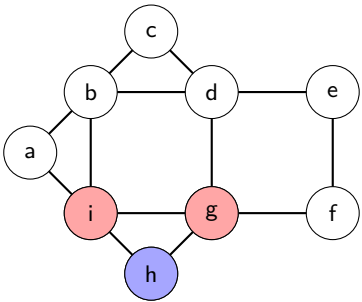
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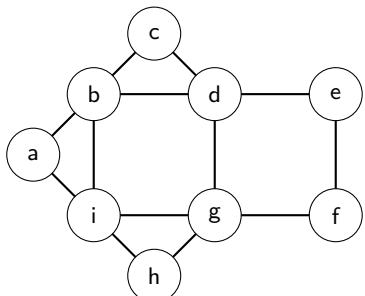
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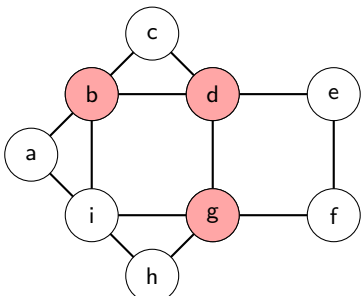
- $I[i, g] = \{i, g, h\}$.
- For $S \subseteq V(G)$, $I[S] = \bigcup_{u,v \in S} I[u, v]$.

P_3-Convex Hull



The P_3-**convex hull** $H(S)$ is the smallest P_3-convex set containing S, obtained by iterating the interval operator until stabilization.

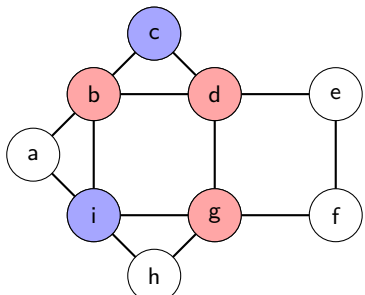
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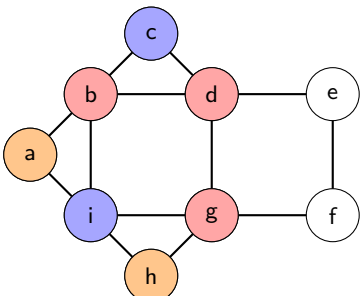
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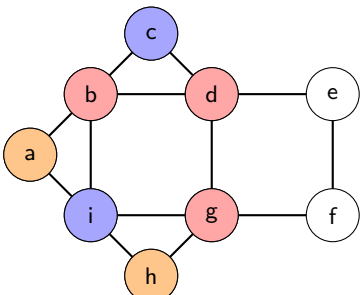
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P_3-Convex Hull

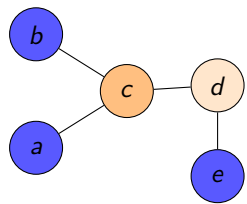


The P_3-**convex hull** $H(S)$ is the smallest P_3-convex set containing S, obtained by iterating the interval operator until stabilization.

- $S = \{b, d, g\}$.
- $I^1[S] = S \cup \{c, i\}$.
- $I^2[S] = I^1[S] \cup \{a, h\}$.
- $I^3[S] = I^2[S]$.
- $H(S) = I^2[S]$.

One round versus repeated activation

- **Interval set:** $\mathcal{I}(S) = V(G)$; minimum size $in(G)$.
- **Hull set:** $\mathcal{I}^t(S) = V(G)$ for some t; minimum size $h(G)$.
- Consequently, $h(G) \leq in(G)$.
- The corresponding decision problems may have different computational complexities.



$S = \{a, b, e\}$, c and d activate in rounds.

Graph products (2019)

Theorems – Coelho, Coelho, Nascimento and Szwarcfiter

For nontrivial connected graphs G, H, ($G \boxtimes H$: strong product.)

$$\boxed{h(G \boxtimes H) = 2.}$$

For nontrivial H and connected nontrivial G, ($G \circ H$: lexicographic product.)

$$\boxed{h(G \circ H) = 2.}$$

For nontrivial connected graphs G, H,

$$\boxed{\max\{h(G), h(H)\} \leq h(G \square H) \leq h(G) + h(H) - 1.}$$

$$\boxed{h(G \square K_n) = \begin{cases} h(G), & \text{if } G \text{ is Type 1,} \\ h(G) + 1, & \text{otherwise.} \end{cases}}$$

Diameter two (2022)

Proposition

If $\operatorname{diam}(G) = 2$ and v is a cut vertex, then v is universal and

$$\boxed{h(G) = \operatorname{cc}(G - v).}$$

Theorem

If G is biconnected and $\operatorname{diam}(G) = 2$, then

$$\boxed{h(G) \leq \lceil \log_2(\Delta(G) + 1) \rceil + 1.}$$

1. Grow a seed set S' until $\mathcal{H}(S')$ dominates G.
2. Each newly selected vertex outside $N[\mathcal{H}(S')]$ causes exponential growth of $|\mathcal{H}(S')|$.
3. One additional vertex suffices to generate the whole graph.

Research direction I: is the bound constant?

Open question posed in 2022

Does there exist an absolute constant C such that every biconnected diameter-two graph G satisfies $h(G) \leq C$?

Stronger conjecture stated in Braully R. Silva's defense

$$\boxed{\text{diam}(G) = 2,\ G \text{ biconnected} \implies h(G) \leq 4.}$$

- Known: the conjectured bound holds for induced-C_6-free graphs.
- Proposed test: search for diameter-two biconnected graphs with $h(G) \geq 5$.

Research direction II: exact complexity

Diameter two

Is there a polynomial-time algorithm to compute $h(G)$ for arbitrary diameter-two graphs?

Diameter three

What is the complexity of computing the P_3-hull number for diameter-three graphs?

- Diameter two: quasi-polynomial exact time is known [2022].
- Diameter four: NP-complete.
- Potential approaches: structural decompositions, parameterized algorithms, and reductions preserving diameter.

Research direction III: Cartesian products

Questions suggested by the 2019 results

- Determine exact values for additional families $G\square H$.
- Identify graph classes where $h(G)$ and $h(H)$ are easy but $h(G\square H)$ is nontrivial.
- Investigate products beyond strong, lexicographic and Cartesian products.

A related parameter: P_3-Carathéodory number (2026)

Definition

The P_3-Carathéodory number $c(G)$ is the least integer k such that, for every $S \subseteq V(G)$ and $v \in \mathcal{H}(S)$, there is $S' \subseteq S$ with $|S'| \leq k$ and $v \in \mathcal{H}(S')$.

For diameter-two graphs:

- If G has a cut vertex, then $c(G) = 2$.
- If G is biconnected and induced-C_6-free, then $c(G) \leq 4$.
- For biconnected G, $c(G) \leq (5 + \sqrt{8\Delta(G) - 15})/2$.

Open question from the 2026 paper

Is $c(G)$ bounded above by an absolute constant for all biconnected diameter-two graphs?

Main results and open questions: diameter two

Graph class	P_3**-hull number** $h(G)$	P_3**-Carathéodory** $c(G)$
Cut vertex v	$\mathrm{cc}(G - v)$	2
Biconnected	$\leq \lceil \log_2(\Delta + 1) \rceil + 1$	$\leq (5 + \sqrt{8\Delta - 15})/2$
Biconnected, induced-C_6-free	≤ 4	≤ 4
Biconnected, general case: constant bound?	OPEN	OPEN
Biconnected, general case: ≤ 4?	OPEN	OPEN
Polynomial-time algorithm for $h(G)$	OPEN	*Not addressed*
Complexity for diameter-three graphs	OPEN	*Not addressed*

Status note: "OPEN" denotes questions explicitly raised in the cited work or proposed research directions, not an exhaustive verification of the literature through October 2026. "Not addressed (2026)" means that the cited Carathéodory paper does not establish a corresponding complexity classification; it does not mean the problem is necessarily open. The $c(G) \leq 4$ row is a suggested strengthening, not a proven universal bound.

Main results: graph products (2019)

Product / condition	**Result for h**
Strong product $G \boxtimes H$	2
Lexicographic product $G \circ H$	2
Cartesian product $G \square H$	$\max\{h(G), h(H)\} \leq h(G \square H) \leq h(G) + h(H) - 1$
Both factors Type 1a	$h(G \square H) \leq h(G) + h(H) - 2$
$G \square K_n$, G Type 1	$h(G)$
$G \square K_n$, G not Type 1	$h(G) + 1$
Exact formulas for arbitrary $G \square H$	OPEN / class-dependent

Discussion: choosing a first problem

1. **Extremal:** prove $h(G) \leq 4$ for a larger diameter-two subclass, or find a counterexample.
2. **Algorithmic:** improve quasi-polynomial exact algorithms for diameter two.
3. **Structural:** characterize useful product factors and special families.

Which direction should we pursue together?

Selected references I

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M. R. Cappelle, E. M. M. Coelho, H. Coelho, B. R. Silva, U. S. Souza, F. Protti. *P_3-convexity on graphs with diameter two: Computing hull and interval numbers*. Discrete Applied Mathematics 321 (2022), 368–378. DOI: 10.1016/j.dam.2022.07.013.

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Thank you!

happy birthday, Luerbio!